■ CHVN Paper Reviewer

Multi-Agent Scientific Paper Review

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File	REVISTA_Project.pdf
Words	6,252
Sections Detected	preamble, introduction, methodology, experiments, results, discussion, conclusions, references, abstract, title, keywords
Review Mode	DEEP
Model	qwen3:8b
Target Publisher	IEEE

Editorial Decision

MAJOR REVISION

Weighted Score: 2.9 / 5.0
Confidence: 85%

Executive Summary

The manuscript presents a solar-powered refrigeration system for food preservation in rural areas, with a focus on thermodynamic analysis and prototype design. However, it lacks sufficient experimental validation, reproducibility, and proper formatting, which are critical for IEEE publication standards.

Decision Rationale

The manuscript has significant technical merit and presents a novel engineering solution for solar-powered refrigeration in rural areas. However, it fails to meet IEEE standards in terms of experimental validation, reproducibility, and reference formatting. These critical issues require major revisions before the manuscript can be considered for publication.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**4.2**	Good	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**3.5**	Good	■■■■■
Statistical Analysis	**4.0**	Good	■■■■■

Results & Evidence	**3.5**	Good	■■■■■
Discussion & Conclusions	**1.5**	Very Weak	■■■■■
Figures & Tables	**2.5**	Weak	■■■■■
Scientific Writing	**2.5**	Weak	■■■■■
References & Citations	**1.0**	Very Weak	■■■■■
Ethics & Transparency	**1.2**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **4.2 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] The title is specific and informative, clearly stating the purpose of the prototype and the technology used.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions.
- [ABSTRACT] The results are quantified with specific values such as COP, volume, and temperature range.
- [KEYWORDS] The keywords are comprehensive and include domain-specific terms for discoverability.

Weaknesses:

- [ABSTRACT] The abstract mentions 'preliminary results' but the results section is incomplete and lacks detailed quantitative data.
- [ABSTRACT] The abstract states the COP is 4.72, which is 75% of the ideal Carnot COP, but the Carnot COP calculation is not provided in the abstract and is only referenced in the discussion section.
- [ABSTRACT] The abstract does not mention the IoT-based monitoring system in detail, which is a key component of the prototype.

Major Comments:

- [ABSTRACT] The abstract claims the prototype has a COP of 4.72, but this value is not supported by a complete thermodynamic analysis in the abstract. The Carnot COP calculation is referenced in the discussion but not included in the abstract, which makes the claim incomplete.
- [ABSTRACT] The abstract states the prototype volume is 286 L, but this value is not clearly supported by the results section, which is incomplete and lacks detailed data.
- [ABSTRACT] The abstract does not mention the IoT-based monitoring system in detail, which is a key component of the prototype and should be highlighted for reproducibility and technical rigor.

Minor Comments:

- [ABSTRACT] The abstract mentions 'preliminary results' but the results section is incomplete and lacks detailed quantitative data.
- [KEYWORDS] The keyword 'R290' is a refrigerant and should be included in the methodology section, not in the keywords.

Recommendations:

- [ABSTRACT] Include the full thermodynamic analysis and Carnot COP calculation in the abstract to support the claim of COP = 4.72.
- [ABSTRACT] Expand the description of the IoT-based monitoring system to highlight its role in real-time data acquisition and system performance.
- [ABSTRACT] Clarify the volume of the prototype and ensure it is supported by detailed data in the results section.
- [KEYWORDS] Remove 'R290' from the keywords and include it in the methodology section for clarity and reproducibility.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 90%

Strengths:

- Title and abstract clearly state the problem of food preservation in rural areas using solar energy.

Weaknesses:

- Introduction lacks a clear statement of the research problem, hypothesis, or objectives.
- Methodology is incomplete and lacks detailed description of the six phases, making reproducibility questionable.

- Experiments section is fragmented and lacks a coherent structure, with missing subsections such as 'Materials and Equipment'.
- Results section is incomplete and lacks sufficient data to support quantitative claims.
- Discussion section contains an equation (COPCarnot) that is not fully defined, and the derivation is missing.
- Conclusions do not clearly align with the stated objectives or results, and lack specific contributions to the field.
- References are incomplete, with missing citations and some references not properly formatted.
- Keywords are missing and not aligned with the content of the manuscript.

Major Comments:

- The manuscript lacks a clear hypothesis or research question, which is essential for a scientific paper.
- The methodology is not fully described, making it impossible to replicate the work.
- The results section is incomplete and lacks sufficient data to support the claims made in the abstract and introduction.
- The discussion section contains an equation that is not properly defined or derived, which undermines the scientific rigor.
- The conclusions do not clearly reflect the results or align with the stated objectives, which is a major flaw in the logical flow.
- The references are incomplete and not properly formatted, which affects the credibility of the manuscript.

Minor Comments:

- The abstract is incomplete and lacks specific details about the methodology or results.
- The title is cut off and does not fully describe the content of the manuscript.
- The keywords section is missing, which is required for indexing and searchability.

Recommendations:

- Complete the introduction by clearly stating the research problem, hypothesis, and objectives.
- Expand the methodology section to include detailed descriptions of each of the six phases, including materials, equipment, and procedures.
- Provide a complete and detailed results section with quantitative data and statistical analysis to support the claims.
- Define and derive all equations in the discussion section, including the COPCarnot equation.
- Revise the conclusions to clearly reflect the results and align with the stated objectives.
- Complete the references section with properly formatted citations and ensure all references are relevant and recent.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

- Clear problem statement and justification for the proposed solution.
- Detailed description of the prototype design and construction process.
- Thermodynamic analysis and comparison with theoretical COP.
- Implementation of an IoT-based monitoring system for real-time data collection.

Weaknesses:

- The 'Experiments' section is incomplete and lacks detailed experimental setup and procedures.
- The 'Results' section is fragmented and lacks a clear structure for presenting the data.
- The methodology is described in a fragmented manner across multiple sections, making it difficult to follow the logical flow.
- The paper lacks a clear description of the experimental setup, including the exact conditions under which the prototype was tested.

Major Comments:

- The 'Experiments' section is incomplete and lacks detailed experimental setup and procedures. This makes it difficult to assess the validity of the results.
- The 'Results' section is fragmented and lacks a clear structure for presenting the data. A more organized presentation of the results would improve clarity and reproducibility.
- The methodology is described in a fragmented manner across multiple sections, making it difficult to follow the logical flow. A more cohesive structure would enhance the readability and reproducibility of the work.

Minor Comments:

- Equation (6) for COPCarnot is not fully defined in the text, though the values are provided.
- The paper lacks a clear description of the experimental setup, including the exact conditions under which the prototype was tested.

Recommendations:

- Expand the 'Experiments' section to include detailed experimental setup and procedures.
- Reorganize the 'Results' section to present the data in a clear and structured manner.
- Ensure that all equations are fully defined and properly introduced in the text.
- Provide a clear description of the experimental setup, including the exact conditions under which the prototype was tested.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.0 / 5.0** | Confidence: 75%

Strengths:

- Engineering design paper: inferential statistical tests are not required for this paper type.
- Measurement uncertainty or variability is acknowledged in the manuscript.
- Operating conditions are described for the reported performance metrics.
- Measurement instruments or data collection methods are mentioned.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- All figures and tables are referenced in the text before they appear, ensuring logical flow.
- Figures and tables are self-explanatory with clear captions and titles that align with the content.
- Equations are numbered sequentially and are introduced with context, making them easy to follow.
- Variables are defined before or immediately after use, ensuring clarity and reducing ambiguity.
- Tables and figures are consistent with the claims made in the text, supporting the scientific narrative.
- The manuscript provides a clear and reproducible methodology, allowing independent researchers to replicate the work.
- The use of IoT and photovoltaic systems is well-documented, with detailed descriptions of components and their roles.
- The comparison of R-290 and R-134a is well-supported by data and references, enhancing the credibility of the claims.
- The thermodynamic analysis is thorough, with clear equations and calculations that align with the theoretical framework.
- The manuscript is well-structured, with clear sections that guide the reader through the research process.

Weaknesses:

- The manuscript does not provide actual data or results from experiments, only theoretical calculations and comparisons. This limits the ability to assess the validity of claims about performance metrics like COP and energy consumption.
- The figures and tables are described in the text, but the actual content of the figures and tables is not provided, making it impossible to verify their accuracy or completeness.
- The equations are referenced in the text, but the actual equations are not included, which is a critical omission for a scientific manuscript.
- The manuscript lacks detailed statistical analysis, such as confidence intervals or p-values, which are necessary for validating the significance of results.
- The manuscript does not provide a clear description of the experimental setup, making it difficult to assess the reproducibility of the work.
- The manuscript does not include a detailed discussion of the limitations of the study, which is essential for a comprehensive evaluation of the research.

Major Comments:

- The manuscript lacks empirical data and experimental results, which are essential for validating the claims made about the performance of the refrigeration system.
- The figures and tables are described in the text, but their actual content is not provided, which is a critical omission for a scientific manuscript.
- The equations are referenced in the text, but the actual equations are not included, which is a critical omission for a scientific manuscript.
- The manuscript lacks detailed statistical analysis, such as confidence intervals or p-values, which are necessary for validating the significance of results.
- The manuscript does not provide a clear description of the experimental setup, making it difficult to assess the reproducibility of the work.

Minor Comments:

- The manuscript could benefit from more detailed descriptions of the IoT system and its integration with the refrigeration system.
- The manuscript could include more information about the specific sensors used and their calibration procedures.
- The manuscript could provide more context about the selection of the R-290 refrigerant, including its advantages and disadvantages in the specific application.

Recommendations:

- Include actual data and results from experiments to support the claims made about the performance of the refrigeration system.
- Provide the actual content of the figures and tables, including their data and visual elements, to ensure transparency and reproducibility.
- Include the actual equations in the manuscript, along with their derivations and explanations, to ensure clarity and correctness.
- Include detailed statistical analysis, such as confidence intervals or p-values, to validate the significance of results.
- Provide a clear description of the experimental setup, including the materials, procedures, and equipment used, to ensure reproducibility.

ResultsReviewer

Results validity and empirical support

Score: **3.5 / 5.0** | Confidence: 70%

Strengths:

- The manuscript provides detailed quantitative results including COP, heat removal rates, and energy consumption metrics.

- The results are presented in tables and figures with clear descriptions and units.
- The comparison of R-290 and R-134a is well-documented with relevant properties and performance metrics.
- The system design and IoT monitoring setup are described with sufficient detail for reproducibility.

Weaknesses:

- The results section lacks a clear distinction between experimental results and theoretical calculations, which could confuse the reader.
- The COP real is claimed to be 75% of the Carnot COP, but no calculation or reference to the Carnot COP under the same conditions is provided.
- The results section does not clearly state whether the COP values were calculated from experimental data or theoretical models.
- The results section does not provide statistical validation or confidence intervals for the COP and other performance metrics.
- The results section does not mention any limitations or uncertainties in the experimental setup or measurements.

Major Comments:

- The claim that the COP real is 4.72 and represents 75% of the Carnot COP is not supported by any detailed calculation or reference to the Carnot COP under the same operating conditions. This is a critical claim that requires validation.
- The results section does not clearly state whether the COP values were calculated from experimental data or theoretical models. This is essential for understanding the validity of the results.
- The results section lacks a clear distinction between experimental results and theoretical calculations, which could confuse the reader and undermine the credibility of the findings.

Minor Comments:

- The results section could benefit from more detailed descriptions of the experimental setup and measurement procedures to enhance reproducibility.
- The results section does not provide statistical validation or confidence intervals for the COP and other performance metrics, which is important for assessing the reliability of the results.
- The results section does not mention any limitations or uncertainties in the experimental setup or measurements, which is important for a complete understanding of the findings.

Recommendations:

- Provide a detailed calculation of the Carnot COP under the same operating conditions to support the claim that the COP real is 75% of the Carnot COP.
- Clarify whether the COP values were calculated from experimental data or theoretical models, and provide the relevant equations and data sources.
- Include statistical validation and confidence intervals for the COP and other performance metrics to assess the reliability of the results.
- Add a section discussing the limitations and uncertainties in the experimental setup and measurements to provide a more complete picture of the findings.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The manuscript provides a clear technical description of the thermodynamic analysis and the COP calculations, which are mathematically consistent and well-defined.
- The discussion includes a comparison of the COP values with other refrigeration technologies (absorption solar and thermoelectric), which is relevant to the field.

Weaknesses:

- The discussion section lacks a comprehensive comparison with prior work, focusing only on citations without detailed analysis.
- The limitations of the study are not clearly stated, which is critical for a rigorous scientific discussion.
- The discussion does not address unexpected findings or anomalies in the data, which is essential for a thorough analysis.
- The theoretical and practical implications of the study are not discussed in sufficient depth.
- The conclusions section is incomplete and does not provide specific, quantified results or future work directions.
- The manuscript does not clearly state how the conclusions are derived from the results, leading to potential overgeneralization.

Major Comments:

- The discussion section fails to provide a detailed comparison with prior work, which is essential for establishing the novelty and significance of the study.
- The limitations of the study are not declared, which is a critical omission for a scientific discussion.
- The discussion does not address unexpected findings or anomalies in the data, which is necessary for a complete analysis.
- The conclusions section is incomplete and does not provide specific, quantified results or future work directions.
- The manuscript does not clearly state how the conclusions are derived from the results, leading to potential overgeneralization.

Minor Comments:

- The discussion section lacks a detailed explanation of the theoretical implications of the findings.
- The conclusions section is not fully developed and does not provide a clear statement of the contribution to the field.

Recommendations:

- The authors should provide a comprehensive comparison with prior work, including detailed analysis of how their results differ from existing studies.
- The authors should clearly state the limitations of the study, such as assumptions made, data constraints, or methodological limitations.
- The authors should address any unexpected findings or anomalies in the data, providing explanations or hypotheses.
- The authors should elaborate on the theoretical and practical implications of their findings for the field of refrigeration technology.
- The authors should complete the conclusions section by providing specific, quantified results and concrete directions for future work.
- The authors should ensure that all conclusions are directly derived from the results and avoid overgeneralization.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical description of the prototype and its components
- Use of specific technical terms and definitions
- Clear alignment between the problem statement and the proposed solution
- Use of relevant references to support claims

Weaknesses:

- Overuse of passive voice
- Lack of specific data and figures to support quantitative claims
- Inconsistent terminology (e.g., 'COP' is used without definition in some sections)

- Unclear methodology for the thermodynamic analysis
- Lack of detailed description of the IoT monitoring system
- Missing figure/table captions and descriptions
- Inconsistent use of units and notation
- Missing statistical analysis and confidence intervals
- Lack of reproducibility details for the prototype
- Missing ethical compliance and conflict-of-interest declaration

Major Comments:

- The manuscript lacks sufficient detail to allow for reproducibility of the prototype. The methodology section is incomplete and lacks specific steps for constructing the prototype.
- The thermodynamic analysis is not clearly described. The equations are not numbered, and the derivation is missing.
- The results section is incomplete and lacks statistical analysis. Confidence intervals and effect sizes are not reported.
- The discussion section is missing and does not address the implications of the findings.
- The conclusion section is missing and does not summarize the key findings or implications of the study.
- The manuscript lacks ethical compliance and conflict-of-interest declaration, which is required for IEEE publication.

Minor Comments:

- The use of passive voice is excessive and should be minimized.
- The manuscript contains several vague quantifiers that need to be replaced with specific numbers.
- The undefined acronyms should be defined on first use.
- The paragraph structure is inconsistent and needs to be improved.
- The use of technical terms should be consistent throughout the manuscript.
- The grammar and syntax are generally correct, but some sentences are run-on and need to be split.

Recommendations:

- Provide detailed steps for constructing the prototype, including materials, tools, and procedures.
- Define all acronyms on first use and ensure consistency throughout the manuscript.
- Include statistical analysis and confidence intervals in the results section.
- Add a discussion section that addresses the implications of the findings.
- Include a conclusion section that summarizes the key findings and implications of the study.
- Add an ethical compliance and conflict-of-interest declaration.

ReferencesReviewer

References quality, recency, and relevance

Score: **1.0 / 5.0** | Confidence: 95%

Strengths:

- Clear alignment between introduction and context of rural electrification and food preservation

Weaknesses:

- References are not properly formatted in IEEE style
- References are URL-only and lack DOIs
- References are not recent or relevant to the technical content of the manuscript
- No seminal works are cited
- No proper citation style is followed

Major Comments:

• The references are not properly formatted in IEEE style. They are URL-only and lack DOIs, which is not acceptable for IEEE journals.

• The references are not recent or relevant to the technical content of the manuscript. They focus on poverty metrics and food waste rather than thermodynamic or IoT standards.

• No seminal works are cited, which is a major omission for a scientific manuscript in a Q1 journal.

• The manuscript lacks proper citation style and formatting, which is a critical issue for IEEE publication standards.

Minor Comments:

• The manuscript does not use 'Tabla' or 'Table' for tables, which is not an issue in this case as no tables are present.

• The manuscript uses 'Cuadro' to label tables. In IEEE publications (including Spanish-language ones), tables must be labeled 'Tabla' or 'Table', not 'Cuadro'.

Recommendations:

• All references must be properly formatted in IEEE style with authors, title, journal, volume, pages, year, and DOI.

• The references must be recent and relevant to the technical content of the manuscript, particularly in thermodynamics, IoT protocols, and photovoltaic systems.

• The manuscript should include seminal works in the field to demonstrate a thorough understanding of the literature.

• The manuscript should be revised to ensure that all citations are in sequential order with no gaps.

• Replace all instances of 'Cuadro' with 'Tabla' (or 'Table') to comply with IEEE style guidelines.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.2 / 5.0** | Confidence: 95%

Strengths:

• The paper presents a clear technical problem and proposes a novel engineering solution for solar-powered refrigeration in rural areas.

• The methodology is structured and follows a logical progression from research to prototype development.

• The discussion includes thermodynamic calculations and comparisons to existing systems, which supports the technical contribution.

Weaknesses:

• The paper lacks transparency in several critical areas, including conflict of interest, funding, AI tool use, and data availability.

• The limitations section is missing, which is essential for responsible scientific reporting.

• The reproducibility of the work is not addressed, making it difficult for independent researchers to replicate the findings.

• The equations in the discussion section are not properly defined, which undermines mathematical rigor.

• The paper does not provide sufficient detail on the experimental setup or parameters used in the prototype development.

Major Comments:

• The paper fails to disclose any use of generative AI tools, which is a mandatory requirement under IEEE/COPE 2024 guidelines.

• There is no conflict of interest declaration, which is required for all journals.

• Funding disclosure is missing, which is essential for transparency.

• The limitations section is entirely absent, which is a major omission in responsible scientific reporting.

• The paper does not provide sufficient detail for reproducibility, making it difficult to replicate the work.

• Equations (6)-(8) are not properly defined or labeled, which violates IEEE standards for mathematical rigor.

Minor Comments:

- The paper lacks a clear statement on data availability, which is important for reproducibility.
- The methodology section does not provide enough detail on the experimental setup or parameters used.
- The conclusion section does not clearly differentiate the novelty of the work from prior IEEE publications.

Recommendations:

- Include a conflict of interest declaration in the manuscript.
- Disclose any funding sources or grants received for this research.
- Declare any use of generative AI tools (e.g., ChatGPT, Copilot) in writing or analysis.
- Add a limitations section that explicitly states the specific constraints and assumptions of the work.
- Provide a data availability statement and detailed experimental parameters for reproducibility.
- Define all equations (6)-(8) with proper units and labels, and ensure they are numbered and referenced correctly.
- Clarify the novelty of the work in relation to prior IEEE publications.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] [ABSTRACT] The abstract claims the prototype has a COP of 4.72, but this value is not supported by a complete thermodynamic analysis in the abstract. The Carnot COP calculation is referenced in the discussion but not included in the abstract, which makes the claim incomplete.

[2] [ABSTRACT] The abstract states the prototype volume is 286 L, but this value is not clearly supported by the results section, which is incomplete and lacks detailed data.

[3] [ABSTRACT] The abstract does not mention the IoT-based monitoring system in detail, which is a key component of the prototype and should be highlighted for reproducibility and technical rigor.

[4] The manuscript lacks a clear hypothesis or research question, which is essential for a scientific paper.

[5] The methodology is not fully described, making it impossible to replicate the work.

[6] The results section is incomplete and lacks sufficient data to support the claims made in the abstract and introduction.

[7] The discussion section contains an equation that is not properly defined or derived, which undermines the scientific rigor.

[8] The conclusions do not clearly reflect the results or align with the stated objectives, which is a major flaw in the logical flow.

[9] The references are incomplete and not properly formatted, which affects the credibility of the manuscript.

[10] The 'Experiments' section is incomplete and lacks detailed experimental setup and procedures. This makes it difficult to assess the validity of the results.

[11] The 'Results' section is fragmented and lacks a clear structure for presenting the data. A more organized presentation of the results would improve clarity and reproducibility.

[12] The methodology is described in a fragmented manner across multiple sections, making it difficult to follow the logical flow. A more cohesive structure would enhance the readability and reproducibility of the work.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [ABSTRACT] The abstract mentions 'preliminary results' but the results section is incomplete and lacks detailed quantitative data.

[2] [KEYWORDS] The keyword 'R290' is a refrigerant and should be included in the methodology section, not in the keywords.

[3] The abstract is incomplete and lacks specific details about the methodology or results.

[4] The title is cut off and does not fully describe the content of the manuscript.

[5] The keywords section is missing, which is required for indexing and searchability.

[6] Equation (6) for COPCarnot is not fully defined in the text, though the values are provided.

[7] The paper lacks a clear description of the experimental setup, including the exact conditions under which the prototype was tested.

[8] The manuscript could benefit from more detailed descriptions of the IoT system and its integration with the refrigeration system.

[9] The manuscript could include more information about the specific sensors used and their calibration procedures.

[10] The manuscript could provide more context about the selection of the R-290 refrigerant, including its advantages and disadvantages in the specific application.

[11] The results section could benefit from more detailed descriptions of the experimental setup and measurement procedures to enhance reproducibility.

[12] The results section does not provide statistical validation or confidence intervals for the COP and other performance metrics, which is important for assessing the reliability of the results.

Specific Improvement Recommendations

1. [ABSTRACT] Include the full thermodynamic analysis and Carnot COP calculation in the abstract to support the claim of COP = 4.72.
2. [ABSTRACT] Expand the description of the IoT-based monitoring system to highlight its role in real-time data acquisition and system performance.
3. [ABSTRACT] Clarify the volume of the prototype and ensure it is supported by detailed data in the results section.
4. [KEYWORDS] Remove 'R290' from the keywords and include it in the methodology section for clarity and reproducibility.
5. Complete the introduction by clearly stating the research problem, hypothesis, and objectives.
6. Expand the methodology section to include detailed descriptions of each of the six phases, including materials, equipment, and procedures.
7. Provide a complete and detailed results section with quantitative data and statistical analysis to support the claims.
8. Define and derive all equations in the discussion section, including the COPCarnot equation.
9. Revise the conclusions to clearly reflect the results and align with the stated objectives.
10. Complete the references section with properly formatted citations and ensure all references are relevant and recent.
11. Expand the 'Experiments' section to include detailed experimental setup and procedures.
12. Reorganize the 'Results' section to present the data in a clear and structured manner.
